If the expansion of the universe reverses, this diffusion of radiation does too. Suddenly the cosmic microwave background, that innocuous low-energy buzz, is blueshifting, rapidly increasing in energy and intensity everywhere, and heading toward very uncomfortable levels.

But that's still not what kills the stars.

It turns out that there is something that can create more high-energy radiation than concentrating the afterglow from space itself being on fire. As the universe has evolved over time, it has taken what was, at the very beginning of the cosmos, a fairly uniform collection of gas and plasma and used gravity to collect that gas into stars and black holes.[fn7] Those stars have been shining for billions of years, sending their radiation out into the void to be dispersed, but not to disappear. Even the black holes have had their chance to shine, producing X-rays as the matter falling into them heats up and creates high-energy particle jets. The radiation produced by stars and black holes is even hotter than the final stages of the Big Bang, and when the universe recollapses, all *that* energy gets condensed too. So rather than being a nicely symmetric process of expansion and cooling followed by coalescence and heating, the collapse is actually *much worse*. If you're ever asked to choose between being at a random point in space just after the Big Bang, or just before the Big Crunch, choose the former.[fn8] The collected radiation from stars and high-energy particle jets, when suddenly condensed and blueshifted to even higher energies by the collapse, will be so intense it will begin to *ignite the surfaces of stars* long before the stars themselves collide. Nuclear explosions tear through stellar atmospheres, ripping apart the stars and filling space with hot plasma.

At this point, things are really very bad. No planet that survived this long could possibly exist un-incinerated when stars themselves are exploded by background light. From here, the intensity of the universe's radiation becomes so high that it can be compared to the central regions of active galactic nuclei, the places where high-energy particles and gamma rays shoot away from supermassive black holes with so much power they make jets of radiation a thousand light-years long. What happens to matter in an environment like that, after it's reduced to its component particles, is uncertain. A collapsing universe will, in the final stages, reach densities and temperatures beyond what we can produce in a laboratory or describe with known particle theories. The interesting question becomes not "Will anything survive?" (because by this point it is very clear that the answer to